

Lavkush Gupta

Machine Learning Engineer

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Profile

Post-graduate with 2 years of experience in executing end-to-end Data Science/Machine Learning projects. Skilled in programming, problem-solving, and innovative thinking. Passionate about learning/developing and deploying impactful models. Eager to enhance professional skills and contribute effectively within a collaborative team environment.

Experience

Machine Learning Engineer - Technology/Data Science

Feb 2023 – present
Mumbai, India

SuperZop

- **Deployed object detection model** on Nvidia Jetson Nano for real-time grain/impurities analysis, ensuring efficient on-device inference with 24-28 FPS.
- **Designed image segmentation model** using advanced deep learning techniques, improving grain commodity detection and dimension measurement accuracy and efficiency.
- **Implemented MLOps principles**, reducing training costs by 60% on Azure Cloud through effective automation and pipeline optimization.
- **Trained an object detection model** for grain detection on **Microsoft Azure Machine Learning**, optimizing performance and reducing costs. Successfully **deployed the model for grain quality analysis** on an **Android app**, providing real-time insights for users.
- **Leveraged cloud technologies** to maintain databases for telemetry data collected from Jetson Nano device and model training, ensuring seamless monitoring and analysis of system performance.
- **Used PowerBI** to create insightful visualizations for tracking grain quality metrics and model performance over time.
- **Integrated and optimized models** within larger systems, achieving exceptional accuracy in controlled laboratory settings.
- **Implemented robust quality control measures** for accurate labeling, ensuring high-quality training data for superior model performance.

Data Science/Technology Intern

Jul 2022 – Jan 2023
Mumbai

SuperZop

- Developed a **color correction model** using **OpenCV** to precisely determine the true colors of objects with reference to custom color card.
- Created an application that **enhances perspective correction** and obtains bird's eye view of objects
- Improved object detection and dimension measurement accuracy by 5% through the implementation of the above-mentioned techniques.

Deep Learning Intern

Sep 2021 – Nov 2021
Gurgaon, India

Helppr.ai

- Developed high-accuracy text classification models for 11 UPSC subjects using Distil BERT.
- Created an image super resolution model based on SRGAN, improving low-resolution images.
- Achieved 95-98% accuracy in various classification models through fine-tuning and error analysis.
- Surpassed organization's goals ahead of schedule with 18% accuracy improvement over previous models.

Education

MSc in Physics

Mumbai, India

NES Ratnam College of Art Science and Commerce. (Affiliated to University of Mumbai)

MSc in Computer Science with a Specialization in Data Science

Mumbai, India

UDCS, University of Mumbai, Mumbai, India

Courses: Python Programming, Statistics, Data Analysis & Visualization, Artificial Intelligence and Machine Learning, SQL & Analytics, Databases Technology, Data Modeling, Research Methodology and Publishing

Highlights: 800+ hours of coursework, 10 coding assignments, 3 projects (Web scraping, EDA, ML)

Skills

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|--|--|---|--|
| • Languages : Python, SQL | • Databases : MySQL, Mongo DB | • Cloud Technologies : Azure Machine Learning Studio, Amazon Cloud Services (AWS) | • Deep Learning Frameworks : PyTorch, Tensorflow, Scikit-learn, Yolo, Detectron2 |
| • Library : Pandas, Numpy, Matplotlib, Seaborn, Sci-kit, Beautiful Soup, RASA for Chatbot, Selenium, NLTK, Streamlit, Flask. | • Other Skills : Presentation, Excel, Tableau, Git, R. | | |

Courses and Certificates

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|---|--|--|---|
| • AWS Practical Data Science Specialization  | • Machine Learning  | • Machine Learning Engineering for Production (MLOps) Specialization  | • Probability and Statistics: To p or not to p?  |
| • Python (Greyatom) | • Deep Learning Specialization  | | |

Languages

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|-----------|---------|-----------|
| • English | • Hindi | • Marathi |
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